

Name: _____ Partner: _____ Period: _____

Part 1 — Graphing $x^2 + y^2 = 36$

1. Substitute each x value into $x^2 + y^2 = 36$ and solve for both y values. Write decimal approximations where needed.

x	y (positive / upper)	y (negative / lower)
-6	_____	_____
$-3\sqrt{2}$	_____	_____
0	_____	_____
$3\sqrt{2}$	_____	_____
6	_____	_____

2. This equation defines _____ explicit function(s). Write them below.

$f_1(x) =$ _____ $f_2(x) =$ _____

3. What is the domain of each explicit function?

Domain of f_1 : _____ Domain of f_2 : _____

Part 2 — Analyzing Co-variation on $x^2 + y^2 = 36$

The table below gives three nearby points on the **upper half** of $x^2 + y^2 = 36$.

x	y
3.0	5.196
3.1	5.155
3.2	5.113

4. Compute $\Delta y / \Delta x$ from $x = 3.0$ to $x = 3.1$:

$$\frac{\Delta y}{\Delta x} = \frac{\text{_____}}{\text{_____}} = \text{_____}$$

5. Compute $\Delta y / \Delta x$ from $x = 3.1$ to $x = 3.2$:

$$\frac{\Delta y}{\Delta x} = \frac{\text{_____}}{\text{_____}} = \text{_____}$$

6. Is $\Delta y / \Delta x$ positive or negative in both cases? _____

7. What does this tell you about how x and y change together along the upper half of the circle?

Part 3 — The Equation $xy = 12$

8. Solve $xy = 12$ for y as an explicit function of x .

$y =$ _____

9. Fill in the table for $x > 0$.

x	y
1	_____
2	_____
3	_____
4	_____
6	_____

10. As x increases (for $x > 0$), does y increase or decrease? _____

11. Compute $\Delta y / \Delta x$ from $x = 2$ to $x = 3$:

$$\frac{\Delta y}{\Delta x} = \frac{\text{_____}}{\text{_____}} = \text{_____}$$

Is this ratio positive or negative? _____ What does this confirm about co-variation?
